14_naive_bayes

May 3, 2026

[ ]:
```python
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns
from IPython.display import display

RANDOM_STATE = 42
sns.set_theme(style="whitegrid")
plt.rcParams.update({"figure.dpi": 120})

DATA_DIR = Path("../data")
if not DATA_DIR.exists():
    DATA_DIR = Path("Durga/Lab/MLDL/data")

pd.set_option("display.max_columns", 120)
```

[2]:
```python
from sklearn.compose import ColumnTransformer
from sklearn.metrics import ConfusionMatrixDisplay, accuracy_score,␣
↪classification_report
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import MultinomialNB
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import OneHotEncoder

df = pd.read_csv(DATA_DIR / "naive_bayes_purchase.csv")
for col in df.columns:
    df[col] = df[col].astype(str).str.strip()
display(df.head())
display(df["Purchase"].value_counts().rename_axis("Purchase").
↪reset_index(name="count"))
```

	Day	Discount	Free Delivery	Purchase
0	Weekday	Yes	Yes	Yes
1	Weekday	Yes	Yes	Yes
2	Weekday	No	No	No
3	Holiday	Yes	Yes	Yes
4	Weekend	Yes	Yes	Yes

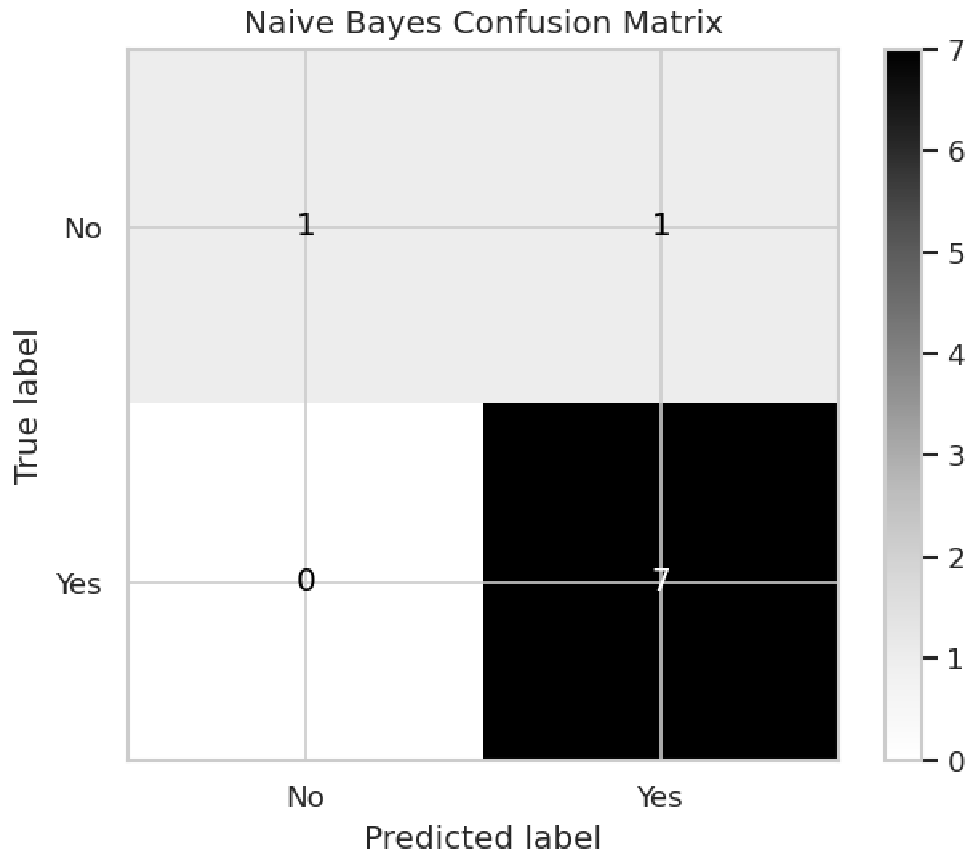
```
  Purchase  count
```

```
0      Yes     24
1       No      6
```

[3]:

```
X = df.drop(columns=["Purchase"])
y = df["Purchase"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=RANDOM_STATE, stratify=y
)

model = Pipeline(
    steps=[
        ("encode", OneHotEncoder(handle_unknown="ignore")),
        ("nb", MultinomialNB(alpha=1.0)),
    ]
)
model.fit(X_train, y_train)
pred = model.predict(X_test)
proba = model.predict_proba(X_test)

print(f"Accuracy: {accuracy_score(y_test, pred):.3f}")
print(classification_report(y_test, pred))
ConfusionMatrixDisplay.from_predictions(y_test, pred, cmap="Greys")
plt.title("Naive Bayes Confusion Matrix")
plt.show()

result = X_test.copy()
result["actual"] = y_test.values
result["predicted"] = pred
for idx, cls in enumerate(model.named_steps["nb"].classes_):
    result[f"prob_{cls}"] = proba[:, idx].round(3)
display(result)
```

```
Accuracy: 0.889
              precision    recall  f1-score   support

          No       1.00      0.50      0.67         2
         Yes       0.88      1.00      0.93         7

    accuracy                           0.89         9
   macro avg       0.94      0.75      0.80         9
weighted avg       0.90      0.89      0.87         9
```



	Day	Discount	Free Delivery	actual	predicted	prob_No	prob_Yes
24	Holiday	No	No	No	No	0.770	0.230
13	Holiday	Yes	Yes	Yes	Yes	0.031	0.969
9	Holiday	Yes	Yes	Yes	Yes	0.031	0.969
22	Holiday	No	Yes	Yes	Yes	0.309	0.691
4	Weekend	Yes	Yes	Yes	Yes	0.013	0.987
8	Weekend	Yes	Yes	Yes	Yes	0.013	0.987
29	Weekend	Yes	Yes	Yes	Yes	0.013	0.987
21	Weekend	Yes	Yes	No	Yes	0.013	0.987
26	Weekday	Yes	Yes	Yes	Yes	0.021	0.979

[4]:
```
feature_names = model.named_steps["encode"].get_feature_names_out(X.columns)
log_probs = pd.DataFrame(
    model.named_steps["nb"].feature_log_prob_,
    index=model.named_steps["nb"].classes_,
    columns=feature_names,
).T
display(log_probs.sort_values(model.named_steps["nb"].classes_[0],␣
 ↪ascending=False))
```

	No	Yes
Discount_No	-1.335001	-2.451005
Free Delivery_No	-1.558145	-2.674149
Day_Holiday	-1.845827	-2.268684
Day_Weekday	-1.845827	-1.863218
Free Delivery_Yes	-2.251292	-1.352393
Day_Weekend	-2.944439	-2.451005
Discount_Yes	-2.944439	-1.421386